Robin Na

Massachusetts Institute of Technology

robin-na.github.io robinna [at] mit.edu robinna

Education

Massachusetts Institute of Technology — **Cambridge, MA**
Ph.D. Candidate in Information Technology — 2022 – Present
- Advisors: Abdullah Almaatouq, Michiel Bakker

Seoul National University — **Seoul, South Korea**
B.S. in Physics & B.A. in Knowledge Ecology (student-designed major) — 2021
- *summa cum laude* | Valedictorian/Dean's Award | Korea Presidential Science Scholar

Research Experience

Amazon — **Palo Alto, CA**
Applied Scientist Intern, Search Experience Team — 2024

Amazon — **Seattle, WA**
Applied Scientist Intern, Science of Science Team — 2023

Carnegie Mellon University — **Pittsburgh, PA**
Research Fellow, Laboratory for Social Minds — 2021 – 2022

New England Complex Systems Institute — **Cambridge, MA**
Undergraduate Researcher, Multiscale Modeling — 2020 – 2021

ASU-Santa Fe Institute Center for Complex Systems — **Tempe, AZ**
Undergraduate Researcher, Collective Logic Lab — 2020

Seoul National University — **Seoul, South Korea**
Undergraduate Researcher, Big Data Studies Lab — 2019
Undergraduate Researcher, Human-Centered Computing Lab — 2019

Korea Advanced Institute of Science and Technology — **Daejeon, South Korea**
Undergraduate Researcher, Complexity and Network Science Group — 2019

Awards and Fellowships (Selected)

- SBS Foundation Graduate Scholarship (3 STEM recipients in Korea) — 2022 – 2028
- DuWayne J. Peterson, Jr. (1955) Fellowship at MIT — 2022 – 2027
- Korea Presidential Science Scholarship in Physics (~20 selected nationally) — 2015 – 2021
- First Place (×3), Seoul National University Research Awards — 2019 – 2021
- Seoul National University Student-Designed Major Scholarship — 2019 – 2021
- Seoul National University Undergraduate Research Grant — 2020 – 2021

Publications

- Work in Progress

1. "Auditing Scientific Literatures with Large Language Models." **Robin Na**, Duncan J. Watts, Abdullah Almaatouq. Working paper.

- Manuscripts Under Review

1. "Designing Behavioral Profiles for Language Model Simulation of Out-of-Distribution Human Behavior." **Robin Na**, Kehang Zhu, Logan Cross, and Michiel Bakker.
   – Under review at *Information Systems Research*.
   – NeurIPS 2026 User Simulation Workshop — **Spotlight**.
2. "Post-training makes large language models less human-like." Marcel Binz et al. including **Robin Na**. Under review at *Science*.
3. "Self-similarity in Pandemic Spread and Fractal Containment Policies." Alexander F. Siegenfeld, Asier Piñeiro Orioli, **Robin Na**, Blake Elias, Yaneer Bar-Yam.

- Peer-Reviewed Publications *Equal Contributions

1. "The Integration of Explanation and Prediction in Behavioral Science" Abdullah Almaatouq, **Robin Na**. *Current Directions in Psychological Science*, 35(3), 2026.
2. "The Diversity of Argument-Making in the Wild: from Assumptions and Definition to Causation and Anecdote in Reddit's 'Change My View.'" **Robin W. Na**, Simon DeDeo. *Proceedings of the* 44th *Annual Conference of the Cognitive Science Society*. 2022.
3. "Snakes or Ladders? Measuring the Intergenerational Performance of Chosŏn's Munkwa Exam Candidates." **Robin W. Na***, Javier Cha*. *International Journal of Korean History*. 2021.
4. "Multi-level Co-authorship Network Analysis on Interdisciplinarity: A Case Study on the Complexity Science Community." **Robin W. Na**, Bongwon Suh. *Proceedings of the* 10th *International Conference on Complex Systems* Springer, Cham. 2020.

Invited Seminars and Conferences (Selected)

– Workshop on Information Systems and Economics	2026
– NeurIPS User Simulation Workshop (**Spotlight**)	2026
– MIT Conference on Digital Experimentation	2024, 2025, 2026
– INFORMS Annual Meeting	2024, 2026
– Symposium on Statistical Challenges in Electronic Commerce Research	2026
– Columbia Business School Workshop on Experimentation for Decision-Making	2026
– MIT Algorithms & Behavior Coffee	2026
– MIT Business Implications of Generative AI Conference	2026
– Conference on Artificial Intelligence, Machine Learning, and Business Analytics	2025
– Wharton AI and the Future of Work Conference	2025
– MIT Behavioral Research Lab Seminar	2025
– Workshop on Information Technologies and Systems	2024
– Stanford Polarization and Social Change Lab Seminar	2024
– AI for Science Workshop at the International Conference on Machine Learning	2024
– International Conference on Computational Social Science	2024
– International Conference on the Science of Science and Innovation	2024
– ACM Collective Intelligence Conference	2024
– Annual Conference of the Cognitive Science Society	2022
– Cultural Evolution Society Conference	2021
– International School and Conference on Network Science	2020
– International Conference on Complex Systems	2020
– Graph Technologies in Digital Humanities — Modelling the Scholarly Process	2020

Research Training Programs

- Deep Learning for Science School at the Lawrence Berkeley National Laboratory	2025
- Santa Fe Institute Complexity-GAINs Summer School on Intelligence & Representation	2023
- Theoretical Organizational Models (TOM) Summer School at Harvard Business School	2023

Teaching Experience

- **Massachusetts Institute of Technology**
  1. Teaching Assistant, Digital Revolution: From Foundations to Future Trends (15.561) Fall 2024, 2025
  2. Teaching Assistant, Action Learning Seminar on Analytics (A-Lab; 15.572) Fall 2023, 2024

Service

- Reviewer: MIT Business Implications on Generative AI Conference, International Conference on Computational Social Science
- Chairperson, Student Education Committee at Seoul National University (2016)

Skills

- **Languages:** Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), R, SQL, Bash
- **Methods:** experimental design, statistical inference, predictive modeling (HTE, distribution shift), time-series forecasting, reinforcement learning, knowledge graphs, network analysis
- **ML Systems:** LLM post-training (Hugging Face/TRL, PEFT/LoRA), inference (vLLM), RAG, Slurm, Git/Linux
- Fluent in English and Korean
  - 21 months of mandatory military service as an English translator at Korea Defense Intelligence Command